

Lifetime and quenching rate constants for Kr2F* and Kr2*

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Citation: Applied Physics Letters **32**, 649 (1978); doi: 10.1063/1.89882
View online: http://dx.doi.org/10.1063/1.89882
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fiber manifested by light-induced refractive-index changes. We have utilized this phenomenon to form high-quality long-length reflection filters in fibers. We have shown that these filters, whose reflectivity can reach nearly 100% of the light launched in the core, can be used as distributed feedback structures to replace the output reflector of an argon-ion laser. The filters have many potential applications, for example, in laser mode control and as synthesized filters with tailored response characteristics for use in high-capacity wavelength-multiplexed light-wave communication systems. As well, the photosensitivity of the fiber guides may find use as a storage mechanism and in wavelength-selective switches and couplers.

The photosensitivity of Ge-doped-core fibers has important implications for the performance of this type of waveguide in a communication system; the dispersion and loss of the guide may be affected in use should the photosensitivity exist in the wavelength regions of interest to optical communications (0.8—0.9, 1.1—1.4 μm). Finally, we note that filters for these spectral regions can be made by difference spatial-frequency techniques using two sources in the visible if adequate photosensitivity is not present in the infrared for direct writing.

The authors are grateful for the technical assistance of R.G. Lamont.

Note added in proof. The maximum reflectivity of an absorption hologram volume recording is approximately 7%, whereas a dielectric hologram volume recording can reach 100% reflectivity. The reflectivities that we observe are therefore not due to an absorption mechanism such as the formation of color centers.

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Lifetime and quenching rate constants for Kr_2F^* and Kr_2^*[a)]

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(Received 13 January 1978; accepted for publication 7 March 1978)

A broadband 2.5-ns-long argon excimer photolytic source was used to initiate the reaction sequence resulting in the formation of Kr_2F^*. The resulting fluorescence signal at 400 nm depended on the radiative and quenching processes of Kr_2^* as well as Kr_2F^*. The radiative lifetime of Kr_2F^* was found to be 181 ± 12 ns. The rate constants for quenching by F_2 and Kr are $4.3 \pm 0.4 \times 10^{-10}$ and $< 2.0 \times 10^{-14}$ $cm^3 s^{-1}$, respectively. The Kr_2^* coupled l_u, O_u^- lifetime is 280 ± 30 ns and the rate constant for quenching of Kr_2^* by F_2 is $2.1 \pm 0.2 \times 10^{-10}$ $cm^3 s^{-1}$.

PACS numbers: 33.70.Fd, 31.70.Hq, 82.20.Rp, 31.50.+w

One of the many interesting results to come out of the rare gas—halide kinetics research has been the discovery of the triatomic rare gas—halide excimer. Intense broadband fluorescence from high-pressure e-beam-excited mixtures of Ar, Kr, and F_2 has been seen at 290 nm [1] and 410 nm.[1,2] The pressure dependence of the fluorescence intensity has been the basis for assigning the ionically bonded excimers Ar_2F^* and Kr_2F^*, respectively, as the sources of this broadband emission.

Recent *ab initio* calculations[3] on Ar_2F^* have reinforced the experimental conclusions. It was shown that the lowest-bound excited state of $Ar_2F(2^2B_2)$ in an isosceles triangle configuration (C_{2v} symmetry) can undergo electric dipole allowed transitions to two lower repulsive covalent states (1^2B_2 and 1^2A_1) with a wavelength near 270 nm and a combined lifetime of 128 ns. Similar calculations on Kr_2F^* have substantiated this species as the 400-nm emitter having a lifetime of 132 ± 25 ns.[4]

The experiment reported here is a direct measurement of the Kr_2F^* lifetime. This was accomplished by optically pumping atomic krypton in the presence of F_2 with a broadband 2.5-ns-long argon excimer source near 129 nm. The broadband excitation source pumps the Kr across the entire absorption width. At the pressures used here, this absorption feature is pressure broadened. The detector views the gas at the center of the cell, which achieves its state of excitation (predominantly) by photoabsorption in the wings of the Lorentzian line shape. At the decay times and pressures measured in this experiment, diffusion losses out of the field of view and wall deactivation are negligible.

The prime advantage of this method is that optical pumping at the wavelength and intensity used here produces no charged particles, a complicating feature of electron or proton beam studies.

The excited atomic krypton (Kr*) thus produced reacts

a)Work done under the auspices of the United States Department of Energy.

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with the krypton and fluorine in the following manner:

$$Kr^* + F_2 \xrightarrow{k_1} KrF^* + F, \quad (1)$$

$$Kr^* + 2Kr \xrightarrow{k_2} Kr_2^* + Kr. \quad (2)$$

The krypton excimer also reacts with the fluorine to give either

$$Kr_2^* + F_2 \xrightarrow{k_3'} Kr_2F^* + F \quad (3)$$

or

$$Kr_2^* + F_2 \xrightarrow{k_3''} KrF^* + F + Kr, \quad (3')$$

$$KrF^* + 2Kr \xrightarrow{k_4} Kr_2F^* + Kr. \quad (4)$$

The relative probability for the accurrence of one route to Kr_2F^* or the other is not known and cannot be discerned in this experiment.

The removal of Kr_2F^* by quenching and radiative processes is given by

$$Kr_2F^* + F_2 \xrightarrow{k_5} 2Kr + F + F_2, \quad (5)$$

$$Kr_2F^* + Kr \xrightarrow{k_6} 3Kr + F, \quad (6)$$

$$Kr_2F^* \xrightarrow{A} h\nu + 2Kr + F. \quad (7)$$

By solving the system of first-order differential equations associated with the reaction mechanism of Table I, one can show that the time dependence of the Kr_2F^* fluorescence signal is given by

$$[Kr_2F^*] = C_1[\exp(-\lambda_1 t) - \exp(-\lambda_2 t)] + C_2[\exp(-\lambda_3 t) - \exp(-\lambda_2 t)] + C_3[\exp(-\lambda_4 t) - \exp(-\lambda_2 t)], \quad (8)$$

where C_1, C_2, and C_3 are constants whose magnitudes and signs depend on the partial pressures of Kr and F_2, the rate constants and the extent of photolytic excitation. The decay frequencies λ_1, λ_2, λ_3, and λ_4 are associated with the removal of Kr_2^*, Kr_2F^*, Kr^*, and KrF^*, respectively. They are given by

$$\lambda_1 = A' + k_3[F_2],$$
$$\lambda_2 = A + k_5[F_2] + k_6[Kr],$$
$$\lambda_3 = k_2[Kr]^2 + k_1[F_2],$$
$$\lambda_4 = A'' + k_7[F_2] + k_4[Kr]^2.$$

TABLE I. Process leading to the formation and decay of Kr_2F^*.

Reaction	Rate constant	Reference
$Kr^* + F_2 \rightarrow KrF^* + F$	$k_1 = 7.2\times10^{-10}$ cm^3 s^{-1}	7
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	$k_2 = 5.5\times10^{-32}$ cm^6 s^{-1}	5
$Kr_2^* + F_2 \rightarrow$ Products	$k_3 = 2.1\times10^{-10}$ cm^3 s^{-1}	This work[a]
$KrF^* + F_2 \rightarrow$ Products	$k_7 = 5.7\times10^{-10}$ cm^3 s^{-1}	6
$KrF^* + 2Kr \rightarrow Kr_2F^* + Kr$	$k_4 = 2.9\times10^{-31}$ cm^6 s^{-1}	6
$Kr_2F^* + F_2 \rightarrow$ Products	$k_5 = 4.3\times10^{-10}$ cm^3 s^{-1}	This work
$Kr_2F^* + Kr \rightarrow$ Products	$k_6 < 2.0\times10^{-14}$ cm^3 s^{-1}	This work
$Kr_2F^* \rightarrow h\nu + 2Kr + F$	$A = 5.52\times10^6$ s^{-1}	This work
$Kr_2^* \rightarrow h\nu' + 2Kr$	$A' = 3.57\times10^6$ s^{-1}	5; this work
$KrF^* \rightarrow h\nu'' + Kr + F$	$A'' = 1.12\times10^8$ s^{-1}	6; 12

[a] $k_3 = k_3' + k_3''$; the relative importance of k_3' and k_3'' cannot be discerned (see text).

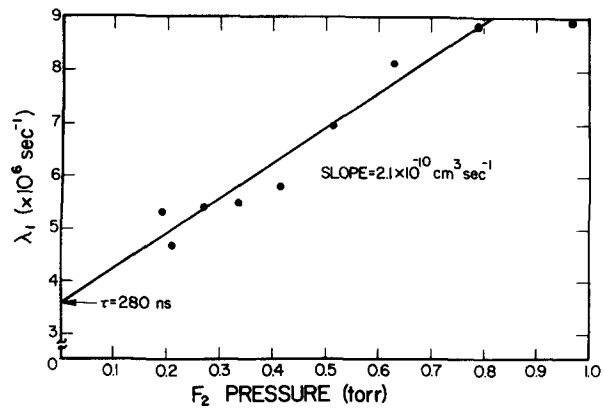


FIG. 1. The decay frequency of the Kr_2F^* fluorescence at 400 nm as a function of F_2 pressure. The mixture is composed of F_2 and Kr with a mole fraction $X_{F_2} = 2\times10^{-4}$.

Using the rate constants of Table I, it can be shown that for the conditions of this experiment the preexponential factors are in the ratio $|C_1| : |C_2| : |C_3| :: 1 : 10^{-3} : 10^{-3}$ and hence all but the first term in Eq. (8) can be neglected. Furthermore, the experimentally determined fluorescence signal has a pure single exponential decay and decomposes with great accuracy into the form $C[\exp(-\lambda_1 t) - \exp(-\lambda_2 t)]$.

The question now is, which of the exponential decay frequencies, λ_1 or λ_2, determines the decay of the Kr_2F^* signal? When $C_1 > 0$, λ_1 determines the decay. When $C_1 < 0$, λ_2 determines the decay. C_1, λ_1, and λ_2 are all pressure dependent and the conditions of the experiment can be adjusted to give either case. Figure 1 shows the plot of the decay frequency of the 400-nm Kr_2F^* signal.[8] The zero pressure intercept is at 3.6×10^6 s^{-1} giving a radiative lifetime of 280 ± 30 ns. This corresponds to the lifetime of Kr_2^* in the coupled l_u, O_u^- electronic states[5] and hence λ_1 determined the decay of the Kr_2F^* signal. The slope of this line is then the quenching of Kr_2^* by F_2 (the quenching of Kr is negligible[5]). Thus $k_3 = 2.1 \pm 0.2\times10^{-10}$ cm^3 sec^{-1}.

This latter number can be justified on the basis of the covalent-ionic curve crossing model[9] (the harpooning mechanism). The ionization potential of Kr_2^* is approximately equal to the ionization potential of $Kr^*(^3P)$. (This is so because the binding energy of Kr_2^* and Kr_2^+ are expected to be very similar.)[10] Thus the ionic curve corresponding to $Kr_2^+F^-$ can be approximated by the ionic curve of $Kr^+(^2P_{3/2})F^-$.[11] The covalent curve of $Kr_2^* + F_2$ at 8.3 eV intersects the latter ionic curve at an internuclear separation of 6.5 Å. The $Kr^*(^3P) + F_2$ covalent curve intersects this same ionic curve at 10.3 Å. Using the known rate of 7.2×10^{-10} cm^3 s^{-1} for the latter process[7] and the fact that the rate constants are proportional to the square of the ionic-covalent crossing radius, one finds that the rate constant for the reaction $Kr_2^* + F_2$ is approximately 2.9×10^{-10} cm^3 s^{-1} which is in good agreement with the measured value considering the crudeness of the model used.

Figures 2 and 3 each show a plot of the decay frequency of the exponential function determining the rising part of the Kr_2F^* fluorescence signal at 400 nm for dif-

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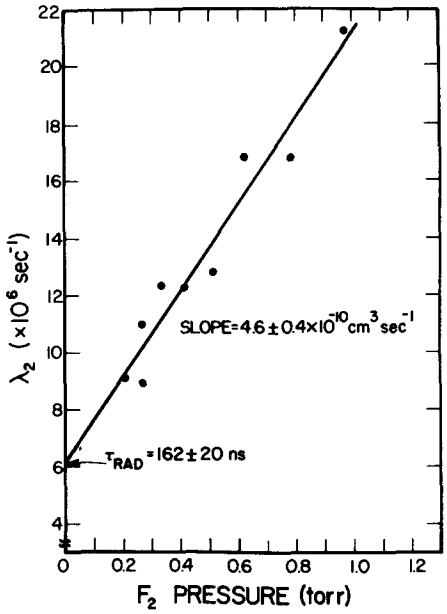


FIG. 2. The decay frequency of the exponential function determining the rising part of the Kr_2F^* fluorescence signal at 400 nm. The mixture is F_2 plus Kr with $X_{F_2} = 2 \times 10^{-4}$.

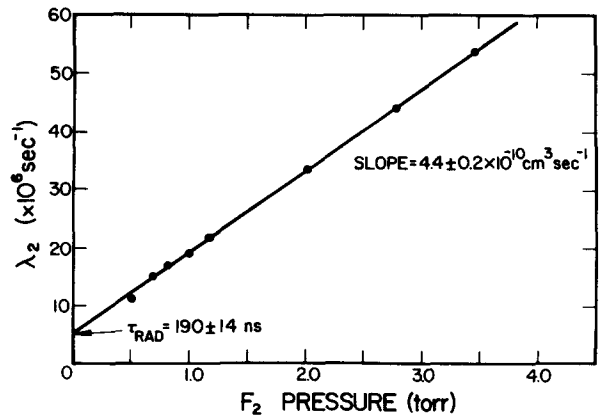


FIG. 3. The decay frequency of the exponential function determining the rising part of the Kr_2F^* fluorescence. Same conditions as in Fig. 2 except that $X_{F_2} = 7 \times 10^{-4}$.

ferent F_2 mole fractions (X_{F_2}). The slope of each is $k_5 + k_6(X_{F_2})^{-1}$ and the intercept is the reciprocal of the radiative lifetime of Kr_2F^*. Solving for the rate constants for quenching of Kr_2F^* by F_2 and Kr, one finds $k_5 = 4.3 \pm 0.4 \times 10^{-10}$ cm^3 s^{-1} and $k_6 < 2.0 \times 10^{-14}$ cm^3 s^{-1}. Taking the proper statistical average of the intercepts, the radiative lifetime for Kr_2F^* becomes 181 ± 12 ns. To the best of our knowledge, these numbers have not previously been reported.[13]

We acknowledge the technical assistance of J.D. King and helpful discussions with Dr. W.R. Wadt.

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